

DEEPIKA GUPTA

Assistant Professor

Dept. of Electronics and Communication Engineering
DSPM International Institute of Technology Naya Raipur (IIIT NR)
E-mail: deepika@iiitnr.edu.in

Research Interest

- Hardware Security System Design based on Advanced Non-Volatile Memories
- Memristor Emulator Design
- 3D Charge Trap NAND Flash Memory
- Advanced MOS Devices
- Digital Circuit and Logic Design

Education

Indian Institute of Technology (IIT) Indore, India.

Doctor of Philosophy (Electrical Engineering)

January 3, 2014- September 21, 2017.

- **Dissertation Title:** Analysis of Charge Trap NAND Flash Memory for Improved Reliability (Defended thesis on 21 September 2017)
- **Research Area:** Design and Analysis of NAND Flash Memory.
- **Advisor:** Dr Santosh Kumar Vishvakarma, Associate Professor, IIT Indore.
- **CGPA:** 9.75/10

Dr B R Ambedkar National Institute of Technology (NIT) Jalandhar, India.

Master of Technology (VLSI Design)

July 2011- September 2013.

- **Dissertation Title:** A New Approach for High Speed & Low Power Feed-Through Logic with Improved Power Delay Product
- **Research Area:** Low Power and High Speed Circuit Design.
- **Advisor:** Dr. Rakesh Kumar Sarin, Professor, NIT Jalandhar.
- **CGPA:** 8.68/10 (Received Gold Medal).

University of Rajasthan, India.

Bachelor of Engineering (Electronics and Communication Engineering)

July 2005-September 2009.

- **Major Project:** Number Plate Recognition System using MATLAB and Neural Network.
- **Score:** 76.4% (first-class).

Board of Secondary Education Rajasthan, India.

Higher Secondary Education

July 2004- May 2005.

- **Score:** 83.5% (first-class with distinction)

Board of Secondary Education Rajasthan, India.

Senior Secondary Education

July 2002- June 2003.

- **Score:** 81% (first-class with distinction).
- Received "Gargi Award" from Govt. of Rajasthan.

Other Qualifications

Graduate Aptitude Test in Engineering (GATE) 2010 and 2013 conducted by MHRD, Govt. of India.

Academic Experience

- Assistant Professor, Grade I in Department of Electronics and Communication Engineering at IIIT Naya Raipur since 1 July 2022.
- Assistant Professor, Grade II in Department of Electronics and Communication Engineering at IIIT Naya Raipur from 5 July 2019 to 30 June 2023.
- Assistant Professor in Veer Surendra Sai University of Technology Burla from 5 September 2018 to 30 April 2019.
- Assistant Professor in University Teaching Departments, Dept. of Electronics and Communication Engineering, Rajasthan Technical University, Kota (Rajasthan) from 4 January 2018 to 1 September 2018.
- Teaching Assistant (TA) at Discipline of Electrical Engineering, IIT Indore from January 2014 to April 2017.
- Assistant Professor in Department of Electronics and Communication Engineering at Mewar University, Chittorgarh (Rajasthan) from July 2013 to December 2013.

- Teaching Assistant (TA) in Department of Electronics and Communication Engineering at NIT Jalandhar from July 2012 to June 2013
- Lecturer in Department of Electronics and Communication Engineering at Modern Institute of Technology and Research Centre (MITRC), Alwar (Rajasthan) from July 2010 to August 2011.

Honours /Awards

- Received **Gold Medal** in M.Tech at Dr. B. R. Ambedkar National Institute of Technology Jalandhar in April 2015.
- Received “Gargi Award “from Govt. of Rajasthan for good academic performance during SSC in January 2004.

List of Publications

Journals:

1. **Deepika Gupta** and S. K. Vishvakarma, “Improved Short Channel Characteristics with Long Data Retention Time in Extremely Short Channel NAND Flash Device”, *IEEE Transactions on Electron Devices*, Vol. 63, No. 2, PP. 668-674, Feb. 2016.
2. **Deepika Gupta** and S. K. Vishvakarma, “Impact of LDD depth Variation on the performance characteristics of SONOS NAND Flash Memory Device”, *IEEE Transactions on Device and Material Reliability*, Vol. 16, No. 3, PP. 298-303, June 2016.
3. **Deepika Gupta** and S. K. Vishvakarma, “Improvement of Short Channel Performance of Junction-free Charge Trapping 3-D NAND Flash Memory”, *IET Micro & Nano Letters*, Vol. 12, No. 1, PP. 64-68, Dec. 2016.
4. S. Chandrakar, **Deepika Gupta**, and M. K. Majumder, “Role of Through Silicon Via in 3D Integration: Impact on Delay and Power”, *Journal of Circuits, Systems and Computers*, World Scientific Publishers, May 2020. (DOI: 10.1142/S0218126621500511)
5. **Deepika Gupta** and S. K. Vishvakarma, “Effect of Spacer Dielectric on the Program Disturb of 3-D Junction-Free NAND Flash Memory”, *Journal of Nanoelectronics and Optoelectronics*, Vol. 14, pp. 1–6, 2019
6. S. Jain, **Deepika Gupta**, S. K. Vishvakarma and V. Neema, “Investigation of BE-SONOS Flash Memory along with Metal Gate and High-k Dielectrics Tunnel Barrier and its Impact in Charge Retention Dynamics,” *Journal of Semiconductors*, Vol. 37, No. 3, Jan. 2016.
7. S. Jain, V. Neema, **Deepika Gupta**, S. K. Vishvakarma, “Investigation of BESONOS Flash Memory with High-k Dielectrics in Tunnel Barrier and its Impact on Charge Retention Dynamics,” *Journal of Nanoelectronics and Optoelectronics* , Vol. 11, No. 6, pp. 663-668, Dec. 2016.
8. **Deepika Gupta**, N. Tiwari, R. K. Sarin, “Analysis of Modified Feed-Through Logic with Improved Power Delay Product”, *International Journal of Computer Application*, Vol. 69, No. 5, pp. 5-8, May 2013.
9. V. Neema, M. Kaur, **Deepika Gupta**, S. K. Vishvakarma, A. Dutt and A. Beohar, “Improvement in Electrical Characteristics of BE-SONOS Using High-k Dielectrics in Tunneling Barrier”, *Transaction on Electrical and Electronic Material*, March 2021. <https://doi.org/10.1007/s42341-021-00307-2>
10. P. K. Sharma, B. Yogesh, **Deepika Gupta** and D. I. Kim, "Performance Analysis of IoT-Based Overlay Satellite-Terrestrial Networks Under Interference," in *IEEE Transactions on Cognitive Communications and Networking*, vol. 7, no. 3, pp. 985-1001, Sept. 2021, doi: 10.1109/TCCN.2021.305250
11. P. K. Sharma, **Deepika Gupta**, and D. I. Kim "Outage performance of 3D mobile UAV caching for hybrid satellite-terrestrial networks," *IEEE Transactions on Vehicular Technology*, IEEE June 2021, DoI: 10.1109/TVT.2021.3089742.
12. P. K. Sharma and **Deepika Gupta**, "Outage performance of multi-UAV relaying-based imperfect hardware hybrid satellite-terrestrial networks," *IEEE Systems Journal*, VOL. 16, NO. 2, June 2022.
13. A. K. Upadhyay, A. K. Kushwaha, **Deepika Gupta**, S. K. Vishvakarma, "Recent Development in Analytical Model for Graphene Field Effect Transistors for RF Circuit Applications", *Journal of Computational Electronics*, Volume 22, No 1, October 2022.
14. S. Chandrakar, **Deepika Gupta**, M. K. Majumder and B. K. Kaushik, "Performance Analysis of Bump in Tapered TSV: Impact on Crosstalk and Power Loss," in *IEEE Open Journal of Nanotechnology*, 2022, doi: 10.1109/OJNANO.2022.3221815.
15. S. Chandrakar, **Deepika Gupta** & Manoj Majumder “Performance analysis of Cu/CNT-based TSV: impact on crosstalk and power”. *Journal of Computational Electronics* **21**, 1262–1274 (2022).
16. S. Chandrakar, **Deepika Gupta** & Manoj Majumder “Impact of TSV Bump and Redistribution Layer on Crosstalk Delay and Power Loss”. *Memories: Materials, devices and Circuits*, **Doi**: <https://doi.org/10.1016/j.memori.2023.100040>

17. **Deepika Gupta**, A. Upadhaya, A. Beohar, S. K. Vishvakarma, "Improvement of Short Channel Effects and Memory Performance of 3-D NAND Flash Memory", *Memories: Materials, Devices and Circuits*, Doi: <https://doi.org/10.1016/j.memori.2023.100031>.
18. N. Bhuwal, M. K. Majumdar, **Deepika Gupta**, "Floating/Grounded Charge Control Memristor Emulator using DVCCTA", *Journal of Computational Electronics*, Vol 23, pp 899-909, 2024.
19. S. Chandrakar, **Deepika Gupta**, and M. K. Majumder, "Role of through silicon via in 3D integration: Impact on delay and power," *Journal of Circuits, Systems and Computers*, World Scientific, vol. 30, no. 3, Mar. 2021, Art. no. 2150051, <https://doi.org/10.1142/S0218126621500511>.
20. S. Chandrakar, **Deepika Gupta**, and M. K. Majumder, "Performance analysis of Cu/CNT-based TSV: impact on crosstalk and power," *Journal of Computational Electronics*, Springer, vol. 21, no. 6, pp. 1262–1274, Dec. 2022, <https://doi.org/10.1007/s10825-022-01937-2>.
21. S. Chandrakar, **Deepika Gupta**, M. K. Majumder and B. K. Kaushik, "Performance Analysis of Bump in Tapered TSV: Impact on Crosstalk and Power Loss," in *IEEE Open Journal of Nanotechnology*, vol. 3, pp. 227–235, 2022, doi: 10.1109/OJNANO.2022.3221815.
22. S. Chandrakar, **Deepika Gupta**, and M. K. Majumder, "Impact of TSV bump and redistribution layer on crosstalk delay and power loss." *Elsevier Memories-Materials, Devices, Circuits and Systems*, vol. 4, July 2023, 100040, <https://doi.org/10.1016/j.memori.2023.100040>.
23. S. Chandrakar, **Deepika Gupta**, M. K. Majumder and B. K. Kaushik, "High-Frequency Electromagnetic Eddy Effect Modeling for Cylindrical and Tapered Bump," in *IEEE Transactions on Electromagnetic Compatibility*, vol. 65, no. 6, pp. 1724–1733, Dec. 2023, doi: 10.1109/TEM.2023.3307816.
24. S. Chandrakar, K. Solanki, **Deepika Gupta** and M. K. Majumder, "Electrical Modeling and Performance Analysis of Cu and CNT Based TSV-Bump-RDL," *IEEE Transactions on Nanotechnology*, vol. 23, pp. 448–455, 2024, doi: 10.1109/TNANO.2024.3408310.
25. T. Kartheek, J. Aditya, **Deepika Gupta**, "Device-Circuit Co-Design of Field-Free VGSOT-MTJ for a Delay-Tunable Energy-Efficient Arbiter PUF", *Journal of Magnetism and Magnetic Material*, Elsevier. (Accepted)
26. T. Kartheek, J. Aditya, **Deepika Gupta**, "A Cascaded Nonlinear VGSOT-MTJ-Based Arbiter Architecture for Variation-Aware VLSI Design", *Integration, The VLSI Journal*, Elsevier. (Accepted)

Conferences:

1. **Deepika Gupta** and S. K. Vishvakarma, "A New Approach to Suppress GIDL in NAND Flash Memory", *18th International Workshop on Physics of Semiconductor Devices (IWPSD)*, Dec 2015, Bangalore, India.
2. M. Kaur, **Deepika Gupta**, V. Neema, S. K. Vishvakarma, V. Vijayvargiya, "Rare Earth Materials and LaON in Tunnel Barrier Engineering With Enhanced Memory Characteristics" *IEEE Prime Asia Conference*, PP. 1-5, Dec 2015, Hyderabad, India.
3. A. Gupta, H. Anwar, **Deepika Gupta**, V. Vijayvargiya and S. K. Vishvakarma, "A Non-Volatile Memory MONOS Device for Improved Stability Applications," *IEEE International Conference on Devices, Circuits and Communications (ICDCCom)*, pp 1-3, Sept. 2014, Ranchi, India.
4. P. K. Sharma, B. Yogesh, and **Deepika Gupta**, "Internet of things-enabled overlay satellite-terrestrial networks in the presence of interference," *National Conference on Communications (NCC 2020)*, IIT Kharagpur, India, Feb. 2020.
5. P. K. Sharma, **Deepika Gupta**, and D. I. Kim, "Cooperative AF-based 3D mobile UAV relaying for hybrid satellite-terrestrial networks," *IEEE Vehicular Technology Conference 2020 (VTC 2020-Spring)*, Antwerp, Belgium, May 2020.
6. K. Das, A. Japa, **Deepika Gupta**, "A Reconfigurable Arbiter PUF based on VGSOT MTJ" *IEEE VLSI Design and Test (VDAT 2022)*, IIT Jammu, India, July 2022.
7. A. Verma, **Deepika Gupta**, and P. K. Sharma, "Outage Performance of Hybrid Satellite-Aerial-Terrestrial Networks in the Presence of Interference", in *2021 Advanced Communication Technologies and Signal Processing (ACTS), IEEE*, NIT Rourkela (Virtual Mode) 2021, 10.1109/ACTS53447.2021.9708132
8. B. Mahapatra, A. Verma, **Deepika Gupta**, P. K. Sharma, and A. K. Turuk, "Traffic-Aware UAV Placement Strategies for Load Balancing in 5G Cellular Hotspots", in *2021 Advanced Communication Technologies and Signal Processing (ACTS), IEEE*, NIT Rourkela (Virtual Mode) 2021, 10.1109/ACTS53447.2021.9708242
9. S. Chandrakar, K. Das, **Deepika Gupta**, M. K. Majumdar, "Signal Integrity and Power Loss Analysis for different Bump Structures in Cylindrical TSV" *IEEE VLSI Design and Test (VDAT 2022)*, IIT Jammu, India, July 2022.
10. S. Chandrakar, **Deepika Gupta** and M. K. Majumder, "Crosstalk and Power Analysis in Tapered based Composite Cu-CNT TSV in 3D IC," *2024 IEEE International Symposium on Circuits and Systems (ISCAS)*, Singapore, Singapore, 2024, pp. 1-5, doi: 10.1109/ISCAS58744.2024.10558624.

11. N. Bhuwal, M.K. Majumdar, **Deepika Gupta**, "Implementation of Floating Charged Memristor Emulator using DVCCTA" 2024 IEEE International Symposium on Circuits and Systems (ISCAS), Singapore, Singapore, 2024, pp. 1-5, doi: 10.1109/ISCAS58744.2024.10558254.
12. N. Bhuwal, K. Solanki, M.K. Majumdar, **Deepika Gupta**, "Flux Controlled Grounded Meminductor Emulator using single DVCCTA" 2024 37th International Conference on VLSI Design and 2024 23 International Conference on Embedded System, Kolkata, India.
13. M. K. Khan, N. Bhuwal, S. Sagar and D. Gupta, "Meminductor Emulator via Flux Approach for Wide Frequency Range Applications," 2024 28th International Symposium on VLSI Design and Test (VDATE), Vellore, India, 2024, pp. 1-6, doi: 10.1109/VDATE63601.2024.10705673.
14. K. Thampula, Samanwaya, A Japa, D Gupta, "Design Insights and Performance Analysis of VGSOT-MTJ for Energy-Efficient Logic and Memory", VLSI SATA, May 2025, Bangalore, India.
15. Mansi Shukla, Anand Singh Thakur, Rohit Chaurasiya and Deepika Gupta, "FPGA Implementation of Resource-Efficient Cube Calculation Architecture using Yavadunam Sutra," 27th International Symposium on VLSI Design and Test (VDATE-2023), In Press, September-2023, BITS-Pilani, Pilani (India). {In Press}
16. Avani Gajallewar, Krishna Agrawal, Arpit Sinha, Deepika Gupta, "Modeling and Side-Channel Attack Resistant VGSOT-MTJ PUF Using ML and DL Analysis", IEEE GCON, IIT Guwahati, 3-5 June 2026.
17. Ananya Rajawat, Palak Maheshwari, Deepika Gupta, "Hardware Accelerator by Implementing Machine Learning Algorithms on FPGA", 2nd International Conference on Emerging Trends in Microelectronics, Communication and Intelligent Systems, MIT Academy of Engineering Pune, 29-30 May 2026.

Book Chapter:

- ❑ P K Sharma, B Yogesh, **Deepika Gupta**, "Overlay Multi-user Satellite Terrestrial network for IoT in the presence of Interference", 5G and Beyond Wireless System, Springer, May 2020.
- ❑ Gupta D. (2020) Low-Voltage Analog Integrated Circuit Design. In: Dhiman R., Chandel R. (eds) Nanoscale VLSI. Energy Systems in Electrical Engineering. Springer, Singapore. https://doi.org/10.1007/978-981-15-7937-0_1

Sponsored Projects

- Reconfigurable and Machine Learning attack resilient spintronic based physically unclonable function, funded by SERB (Sanctioned on 8 May 2023) under SURE scheme.
- Design and Analysis of Memristor based Nonvolatile SRAM for Low Power Resource Constraint IoT Application, Funded By CSIR (Sanctioned on 18 June 2024) under ASPIRE scheme.
- Sanctioned total of \$1000 to conduct events by IEEE

Patents:

- Deepraj Chowdhury, Maifuz Ali Deepika Gupta, Smart Traffic Signal Management System for Roadways, Awarded on 17 August 2021.
- Deepraj Chowdhury, Lakhindar Murmu, Deepika Gupta, Artificial Intelligence Controlled Door Access System For Dwellings, Awarded on 9 November 2022

Organisational Activities

- Working as the Chairperson of IEEE Activities at IIIT Naya Raipur
- Worked as Time Table Coordinator at IIIT Naya Raipur.
- Worked as TA duty Coordinator at IIIT Naya Raipur.
- Worked as warden for Girls Hostel at IIIT Naya Raipur.
- Worked as Organizing Secretary at 8th IEEE Conference WIECON.
- Worked as Organizing Secretary at 2nd IEEE Conference ICBDS.
- Coordinator to Conduct institutional activity under Institutional Social Responsibility Cell of IIIT NR.

(DEEPIKA GUPTA)